

Lesson purpose

Students learn to interpret and draw molecule models for composition, chemical change, and temperature-related changes in motion and spacing.

1. Intro Molecules

Open by telling students the day is about making invisible particles visible with models. The target has three pieces: draw molecules, show before/after a chemical change, and show what temperature changes do.

2. What a molecule model shows

Emphasize that the colors and letters are a code. Students should not worry that real atoms look exactly like circles.

3. Read the code in a model

Use the repeated sentence frame: what stays the same, and what changes?

4. Before and after: count atoms

Connect to conservation of matter at the atom-count level. Students do not need coefficients yet; the visual count is enough for grade 7.

5. Chemical change = new particle arrangement

A chemical change creates new substances because atoms are connected differently. Temperature affects motion/spacing unless chemistry also occurs.

6. Teacher model: how to draw a chemical change

Model the exact student process: make a key, draw before, draw after, count both sides.

7. Formative check 1

Expected response: Model B is acceptable for conservation of atom counts; Model A loses one red atom. Push students to name model evidence.

8. Temperature changes molecular motion

Use the hot/cold water idea: hot water and food coloring mix faster because molecules move faster.

9. How to draw temperature in a model

Motion lines are model symbols. They show speed in a still drawing.

10. Compare the two model stories

This is the conceptual hinge: chemical rearrangement versus temperature motion/spacing.

11. Formative check 2

Circulate for common errors: drawing single atoms instead of H_2/O_2 and losing atoms during the redraw.

12. Exit ticket

Collect or spot-check for whether students can separate chemical rearrangement from temperature motion/spacing.

Source notes

Content accuracy checked against OpenStax Chemistry: Atoms First 2e, sections 1.1 and 1 summary, and ACS Middle School Chemistry Lesson 1.2 Molecules in Motion, last modified 2024-07-18.